

Class 18: Pertussis Mini-Project

Tonhi Dinh (A18508020)

Table of contents

Background	1
CDC tracking data	1
Exploring CMI-PB data	3

Background

Pertusis (whooping cough) is a common lung infection caused by the bacteria B. Pertussis. It can infect everyone but is most deadly for infants (under 1 year of age)

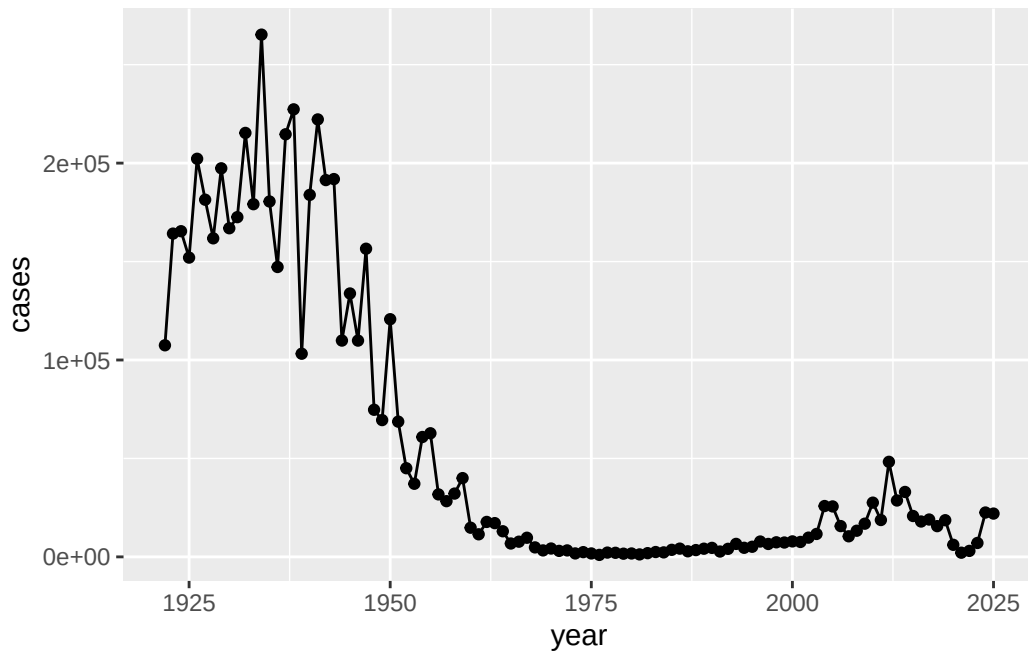
CDC tracking data

The CDC tracks the number of Pertussis cases:

I want a plot of year vs. cases

```
library(ggplot2)

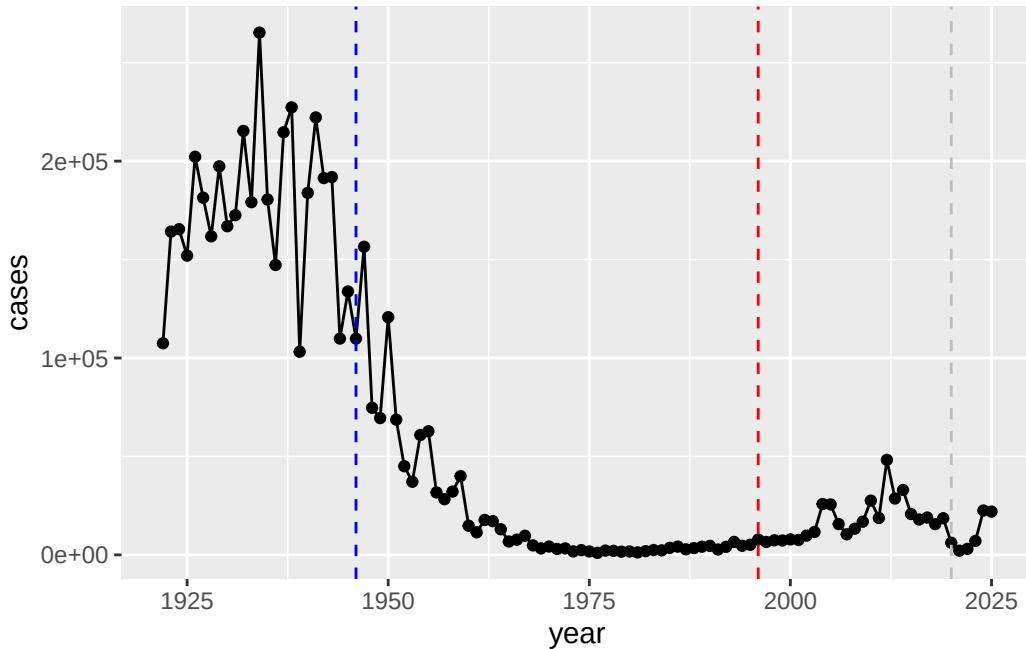
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```



Q. Add annotation lines for the major milestones of wP vaccination roll-out (1946) and the switch to the aP vaccine (1996)

```
library(ggplot2)

ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col = "blue", lty = 2) +
  geom_vline(xintercept = 1996, col = "red", lty = 2) +
  geom_vline(xintercept = 2020, col = "grey", lty = 2)
```



It is clear from the CDC data that pertussis cases are once again increasing. For example, we can see that in 2012 the CDC reported 48,277 cases of pertussis in the United States. This is the largest number of cases reported since 1955, when 62,786 cases were reported. The pertussis field has several hypotheses for the resurgence of pertussis including (in no particular order): 1) more sensitive PCR-based testing, 2) vaccination hesitancy 3) bacterial evolution (escape from vaccine immunity), 4) waning of immunity in adolescents originally primed as infants with the newer aP vaccine as compared to the older wP vaccine.

Exploring CMI-PB data

The CMI-PB project < <https://www.cmi-pb.org> > mission is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination

Basically, make available a large data set on the immune response to Pertussis. They use a “booster” vaccination as a proxy for Pertussis vaccination

They make their data available as JSON format API. We can read this into R with the `read_json()` function from the `jsonlite` package:

```
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many aP and wP individuals are in this subject table?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

How many male/female are there?

```
table(subject$biological_sex)
```

```
Female  Male
112     60
```

Q. What is the breakdown of biological_gender and race?

Q. Is this representative of the US population?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12

Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

We can read more tables from the CMI-PB database

```
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen",
                      simplifyVector = TRUE)
ab_titter <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer",
                      simplifyVector = TRUE)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

```
head(ab_titter)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
--	------	--------------------------

1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

To make sense of all this data, we need to “join” (a.k.a. “merge” or “link”) all these tables together. Only then will you now that a given Ab measurement (from the `ab_titter` table) was collected on a certain date (from the `specimen` table) from a certain wP or aP subject (from the `subject` table).

We can use **dplyr** and the `*_join()` family of functions to do this

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- inner_join(subject, specimen)
```

Joining with ``by = join_by(subject_id)``

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

Let's do one more `inner_join()` to join the `ab_titter` with all our meta data

```
abdata <- inner_join(ab_titter, meta)
```

Joining with ``by = join_by(specimen_id)``

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	UG/ML	2.096133	1	wP	Female
2	IU/ML	29.170000	1	wP	Female
3	IU/ML	0.530000	1	wP	Female

4	IU/ML	6.205949	1	wP	Female
5	IU/ML	4.679535	1	wP	Female
6	IU/ML	2.816431	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	-3	0	Blood
3	-3	0	Blood
4	-3	0	Blood
5	-3	0	Blood
6	-3	0	Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

Q. How many different Ab “isotypes” values are in this dataset?

```
table(abdata$isotype)
```

IgE	IgG	IgG1	IgG2	IgG3	IgG4
6698	7265	11993	12000	12000	12000

Q. How many different “antigen” values are measured

```
table(abdata$antigen)
```

ACT	BETV1	DT	FELD1	FHA	FIM2/3	LOLP1	LOS	Measles	OVA
1970	1970	6318	1970	6712	6318	1970	1970	1970	6318
PD1	PRN	PT	PTM	Total	TT				
1970	6712	6712	1970	788	6318				

Let's focus on IgG isotype

```
igg <- abdata |>
  filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	IU/ML	0.530000	1	wP	Female
2	IU/ML	6.205949	1	wP	Female
3	IU/ML	4.679535	1	wP	Female
4	IU/ML	0.530000	3	wP	Female
5	IU/ML	6.205949	3	wP	Female
6	IU/ML	4.679535	3	wP	Female

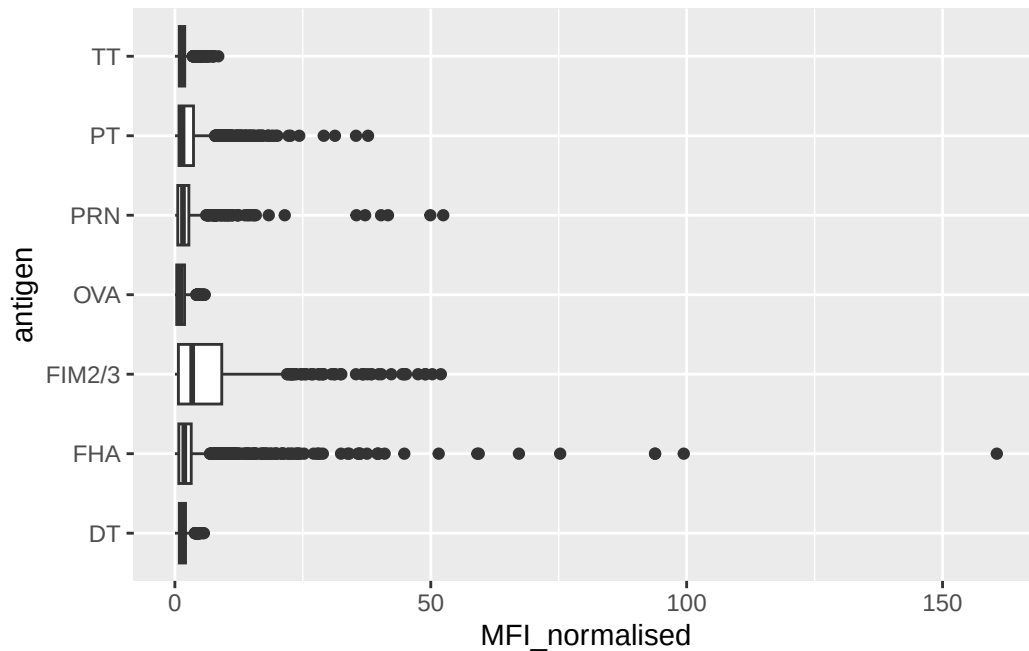
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	-3	0	Blood
3	-3	0	Blood
4	-3	0	Blood
5	-3	0	Blood
6	-3	0	Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

Make a plot of “MFI_normalised” values for all **antigen** values.

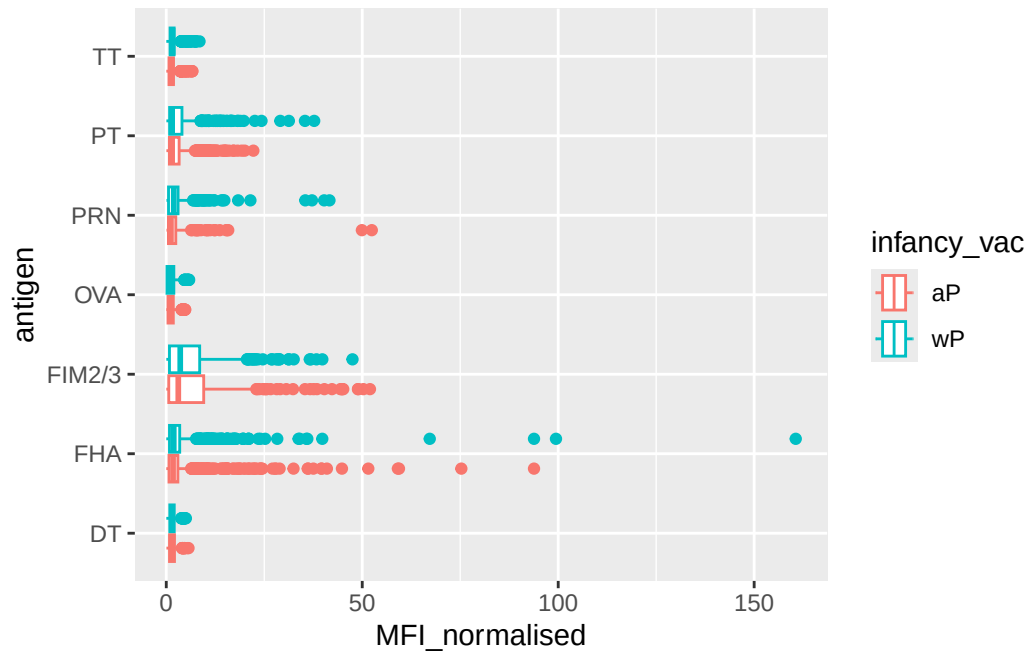
```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



The antigen “PT”, “FIM2/3” and “FHA” appear to have the widest range of values.

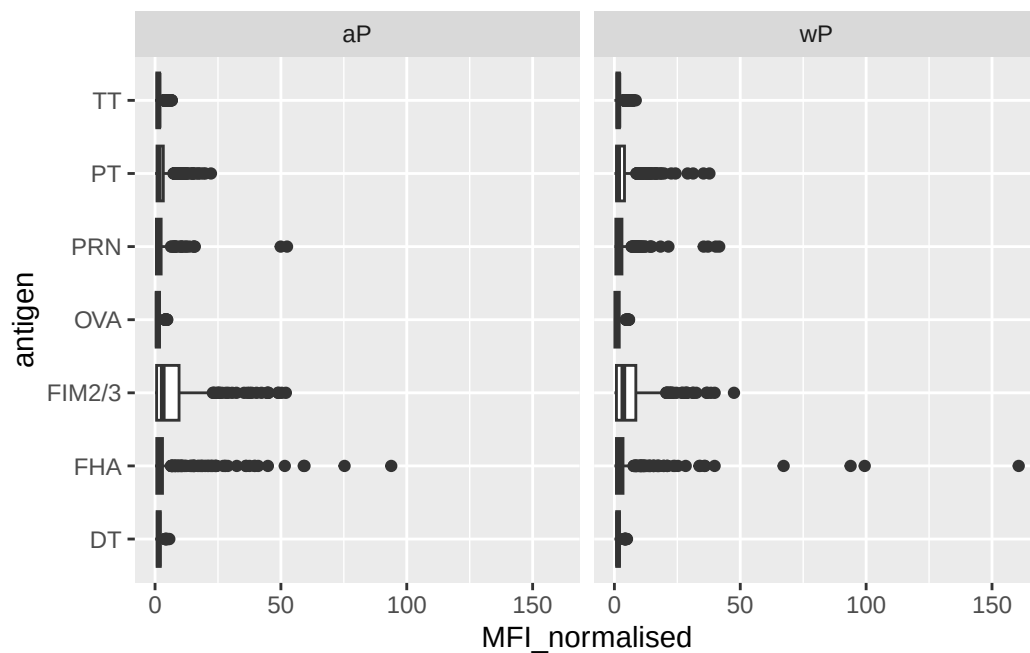
Q. Is there a difference for these responses between aP and wP individuals?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



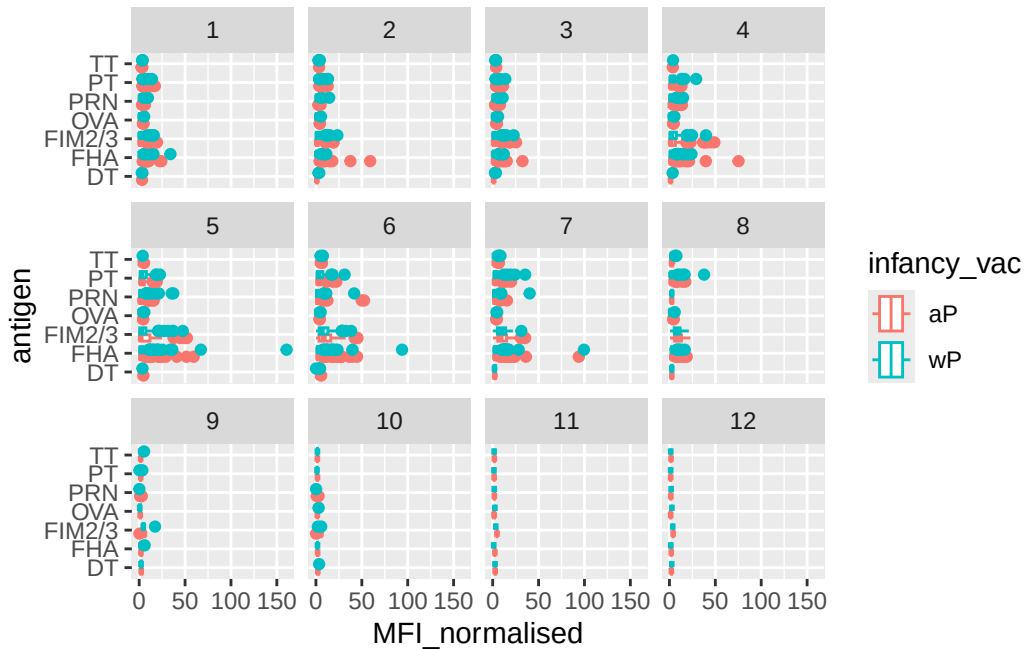
```
library(ggplot2)

ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



Is there a difference with time (i.e. before booster shoot vs after booster shot)

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~visit)
```



Lets finish this section by looking at the 2021 dataset IgG PT antigen levels time-course:

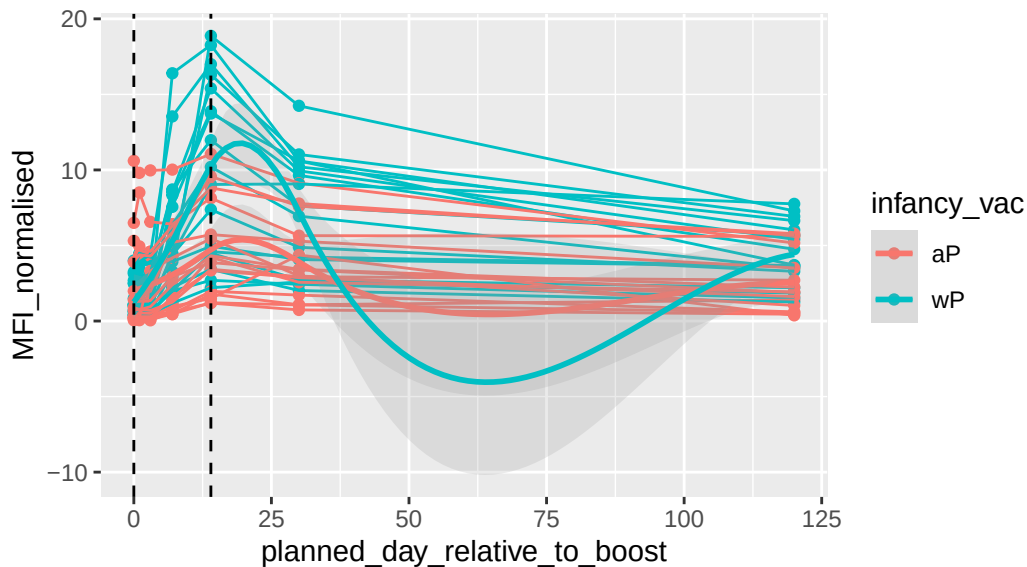
```
## Filter to 2021 dataset, IgG and PT only
ab.PT.21 <- abdata |> filter(dataset == "2021_dataset",
                             isotype == "IgG",
                             antigen == "PT")

ggplot(ab.PT.21) +
  aes(x=planned_day_relative_to_boost,
       y=MFI_normalised,
       col=infancy_vac,
       group=subject_id) +
  geom_point() +
  geom_line() +
  geom_smooth(aes(group = infancy_vac),
              alpha = 0.2) +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



```
trajectory_data <- ab.PT.21 %>%
  group_by(infancy_vac, planned_day_relative_to_boost) %>%
  summarise(mean_MFI = mean(MFI_normalised, na.rm = TRUE))
```

`summarise()` has regrouped the output.

i Summaries were computed grouped by infancy_vac and planned_day_relative_to_boost.

i Output is grouped by infancy_vac.

i Use `summarise(.groups = "drop\_last")` to silence this message.

i Use `summarise(.by = c(infancy\_vac, planned\_day\_relative\_to\_boost))` for per-operation grouping (`?dplyr::dplyr\_by`) instead.

```
ggplot() +

# individual subjects
geom_line(data = ab.PT.21,
  aes(x = planned_day_relative_to_boost,
    y = MFI_normalised,
    group = subject_id,
    color = infancy_vac),
  alpha = 0.3) +
```

```
# mean trajectory
geom_line(data = trajectory_data,
          aes(x = planned_day_relative_to_boost,
              y = mean_MFI,
              color = infancy_vac,
              group = infancy_vac),
          size = 1.5) +

geom_point(data = trajectory_data,
           aes(x = planned_day_relative_to_boost,
               y = mean_MFI,
               color = infancy_vac),
           size = 3) +

labs(x = "Days Relative to Boost",
     y = "Normalized MFI",
     color = "Infancy Vaccine") +

theme_minimal()
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.

